

Endpoint products in distributional fractional Sobolev spaces: a strict-loss theorem and a local Hardy–Sobolev subcase

Candidate partial result for arXiv:2111.13942

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Status. Candidate substantial partial result, likely valid; human review is needed. The exact endpoint algebra question remains open.

1 Source question

Comi and Stefani ask in Remark 4.6 of *Leibniz rules and Gauss–Green formulas in distributional fractional spaces* [1] whether, for $0 < \alpha < 1$,

$$f, g \in S^{\alpha,1}(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n) \implies \nabla_{\text{NL},w}^\alpha(f, g) \in L^1(\mathbb{R}^n; \mathbb{R}^n).$$

The same remark proves the exact equivalence

$$\nabla_{\text{NL},w}^\alpha(f, g) \in L^1 \iff fg \in S^{\alpha,1}(\mathbb{R}^n). \quad (1.1)$$

Thus the question is precisely whether $S^{\alpha,1} \cap L^\infty$ is an algebra.

Here $S^{\alpha,1}$ consists of $u \in L^1$ whose distributional Riesz fractional gradient $\nabla^\alpha u$ belongs to L^1 . We use

$$[u]_{W^{\beta,1}} := \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \frac{|u(x+h) - u(x)|}{|h|^{n+\beta}} dx dh. \quad (1.2)$$

2 Main partial result

Theorem 1 (Every strict lower order is stable). *Let $0 < \beta < \alpha < 1$ and $f, g \in S^{\alpha,1}(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$. Then*

$$f, g, fg \in W^{\beta,1}(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n) \subset S^{\beta,1}(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n).$$

Set

$$\mathcal{A}_{\alpha,\beta}(u) := \frac{\gamma_{n,\alpha}}{\alpha - \beta} \|\nabla^\alpha u\|_{L^1} + \frac{2}{\beta} \|u\|_{L^1}, \quad (2.1)$$

where $\gamma_{n,\alpha}$ is the translation constant in [2, Proposition A.1]. Then

$$[fg]_{W^{\beta,1}} \leq n\omega_n (\|g\|_{L^\infty} \mathcal{A}_{\alpha,\beta}(f) + \|f\|_{L^\infty} \mathcal{A}_{\alpha,\beta}(g)), \quad (2.2)$$

$$\left\| \nabla^\beta(fg) \right\|_{L^1} \leq \mu_{n,\beta} n\omega_n (\|g\|_{L^\infty} \mathcal{A}_{\alpha,\beta}(f) + \|f\|_{L^\infty} \mathcal{A}_{\alpha,\beta}(g)). \quad (2.3)$$

Consequently $\nabla_{\text{NL},w}^\beta(f, g) \in L^1$ exists and

$$\nabla^\beta(fg) = g\nabla^\beta f + f\nabla^\beta g + \nabla_{\text{NL},w}^\beta(f, g) \quad \text{in } L^1. \quad (2.4)$$

Theorem 2 (Exact-order local Hardy–Sobolev subcase). *Let $0 < \alpha < 1$ and let*

$$hS^{\alpha,1}(\mathbb{R}^n) := F_{1,2}^\alpha(\mathbb{R}^n)$$

be the inhomogeneous Triebel–Lizorkin, or local Hardy–Sobolev, space. Then

$$F_{1,2}^\alpha(\mathbb{R}^n) \hookrightarrow S^{\alpha,1}(\mathbb{R}^n), \quad (2.5)$$

and $F_{1,2}^\alpha \cap L^\infty$ is an algebra. In particular, if $f, g \in F_{1,2}^\alpha \cap L^\infty$, then the exact-order implication in (1.1) holds. More quantitatively,

$$\|fg\|_{F_{1,2}^\alpha} \leq C(\|f\|_{F_{1,2}^\alpha} \|g\|_{L^\infty} + \|g\|_{F_{1,2}^\alpha} \|f\|_{L^\infty}). \quad (2.6)$$

This local subclass strictly contains the global Hardy–Sobolev space $(I - \Delta)^{-\alpha/2} H^1$ used in [2].

3 Idea of the proof

The full hypotheses already give an endpoint translation modulus: $S^{\alpha,1}$ embeds into $B_{1,\infty}^\alpha$. Integrating that modulus against $|h|^{-n-\beta}$ is finite at every $\beta < \alpha$, and the elementary product difference inequality preserves it. This proves the full-class strict-loss theorem without invoking an endpoint Riesz-transform estimate.

At the exact order, the smaller local Hardy–Sobolev scale has precisely the Littlewood–Paley cancellation needed for pointwise multiplication. Sickel’s multilinear theorem supplies the algebra estimate. A low/high-frequency split then shows that its fractional gradient is integrable, placing the product inside the source’s $S^{\alpha,1}$ space.

The open endpoint is not obtained by taking $\beta \uparrow \alpha$: the bound has the sharp visible loss $(\alpha - \beta)^{-1}$, and exact $B_{1,\infty}^\alpha$ regularity alone does not recover the missing Riesz-transform cancellation.

4 Proofs

Proof of Theorem 1. The translation estimate of Bruè–Calzi–Comi–Stefani [2, Proposition A.1], at $p = 1$, is

$$\|\tau_h u - u\|_{L^1} \leq \gamma_{n,\alpha} |h|^\alpha \|\nabla^\alpha u\|_{L^1}. \quad (4.1)$$

For $|h| > 1$ we also have $\|\tau_h u - u\|_{L^1} \leq 2\|u\|_{L^1}$. Polar integration gives

$$\begin{aligned} [u]_{W^{\beta,1}} &= \int_{\mathbb{R}^n} \frac{\|\tau_h u - u\|_{L^1}}{|h|^{n+\beta}} dh \\ &\leq n\omega_n \left(\frac{\gamma_{n,\alpha}}{\alpha - \beta} \|\nabla^\alpha u\|_{L^1} + \frac{2}{\beta} \|u\|_{L^1} \right) = n\omega_n \mathcal{A}_{\alpha,\beta}(u). \end{aligned} \quad (4.2)$$

Pointwise,

$$|(fg)(x+h) - (fg)(x)| \leq |g(x+h)| |f(x+h) - f(x)| + |f(x)| |g(x+h) - g(x)|.$$

After integration in x and h , (4.2) gives (2.2). The singular-integral definition is absolutely convergent for $u \in W^{\beta,1}$ and yields

$$\left\| \nabla^\beta u \right\|_{L^1} \leq \mu_{n,\beta} [u]_{W^{\beta,1}}, \quad (4.3)$$

which proves (2.3) and the asserted inclusions. Applying the same argument to each input places f, g in $S^{\beta,1} \cap L^\infty$. Now the equivalence (1.1), with α replaced by β , gives (2.4). $\square$

Proof of Theorem 2. Sickel's Theorem 1 [3], with

$$m = 2, \quad p_1 = p_2 = p = 1, \quad q = 2, \quad s = \alpha,$$

applies because

$$\alpha > n \max(0, 1 - 1/2 - 1) = 0, \quad \max_j(1/p_j) = 1 \leq 1/p = 1 \leq \sum_j 1/p_j = 2.$$

It gives (2.6), hence the algebra assertion.

It remains to prove (2.5). Let $J = (I - \Delta)^{1/2}$. The lifting property gives $h := J^\alpha f \in F_{1,2}^0 = h^1$, the local Hardy space. Since $h^1 \hookrightarrow L^1$ and the Bessel kernel of $J^{-\alpha}$ is in L^1 , we obtain $f = J^{-\alpha} h \in L^1$.

For the j th fractional-gradient component,

$$\nabla_j^\alpha f = T_j h, \quad \widehat{T_j h}(\xi) = m_j(\xi) \widehat{h}(\xi), \quad m_j(\xi) = i \xi_j |\xi|^{\alpha-1} (1 + |\xi|^2)^{-\alpha/2}. \quad (4.4)$$

Choose a smooth cutoff χ supported near the origin. The inverse Fourier transform of χm_j is integrable: it is bounded locally and, being a smooth cutoff of a degree- α homogeneous symbol that vanishes at the origin, decays as $O(|x|^{-n-\alpha})$. Thus the low-frequency operator maps L^1 to L^1 . The symbol $(1 - \chi)m_j$ is a standard order-zero multiplier supported away from the origin, so the associated local Calderón–Zygmund operator is bounded on h^1 . Since $h^1 \hookrightarrow L^1$, both pieces of $T_j h$ lie in L^1 . Hence $\nabla_j^\alpha f \in L^1$, proving (2.5).

Finally $H^1 \hookrightarrow h^1$, so the global space embeds in the local one. The inclusion is strict: a nonzero nonnegative C_c^∞ function belongs to $F_{1,2}^\alpha$, while it cannot belong to $(I - \Delta)^{-\alpha/2} H^1$ because every global H^1 function has integral zero and J^α preserves the zero-frequency value. $\square$

5 Verification and limitations

The proof has three independent consistency checks. First, the constants in (4.2) are exactly the radial integrals $\int_0^1 r^{\alpha-\beta-1} dr$ and $\int_1^\infty r^{-\beta-1} dr$. Second, the product estimate uses only translations and L^∞ bounds, so no hidden L^1 Riesz-transform estimate enters Theorem 1. Third, Sickel's parameter conditions include $p = 1$, $q = 2$, and every $s = \alpha > 0$ exactly as substituted above.

The theorem does *not* establish $fg \in S^{\alpha,1}$ for arbitrary bounded $S^{\alpha,1}$ inputs. Nor does it assert that the global Hardy–Sobolev space is an algebra: if $0 \neq f$ is real valued there, then $\int f^2 > 0$, whereas every member of that global space has integral zero. What is proved is that bounded products from the global space land in the larger local space $F_{1,2}^\alpha$, and hence in $S^{\alpha,1}$.

Eight focused upgrade attempts also tested direct endpoint Kato–Ponce estimates, the known fractional chain-rule counterexample, accumulating-jump and lacunary constructions, a closed-graph norm blow-up, and the $\alpha = 0$ obstruction. None produced a valid full proof or counterexample. The detailed audit is in the associated attempt note.

Cheap run indexes and bounded searches through 12 August 2026 for the arXiv id, quoted question, endpoint Leibniz/Kato–Ponce formulations, and the $S^{\alpha,1}$ algebra problem found no later paper explicitly resolving the full question. The local Hardy–Sobolev algebra estimate is known literature [3]; its implication for the source problem is identified here. The full-class strict-loss theorem is the promoted new deduction.

6 Human-review recommendation

Review (i) the normalization of the double-difference seminorm in (4.2), and (ii) the low-frequency kernel assertion in (4.4). If those standard points are accepted, Theorem 1 and the exact-order

subclass theorem are ready to retain as a substantial partial result. The packet should not be upgraded to a full solution without a new endpoint cancellation argument or an explicit counterexample.

References

- [1] G. E. Comi and G. Stefani, *Leibniz rules and Gauss–Green formulas in distributional fractional spaces*, Rev. Mat. Iberoam. 39 (2023), 1289–1330; arXiv:2111.13942.
- [2] E. Bruè, M. Calzi, G. E. Comi, and G. Stefani, *A distributional approach to fractional Sobolev spaces and fractional variation: asymptotics II*, C. R. Math. Acad. Sci. Paris 360 (2022), 589–626; arXiv:2011.03928.
- [3] W. Sickel, *Pointwise multiplication in Triebel–Lizorkin spaces*, Forum Math. 5 (1993), 73–92, Theorem 1.

$$\lim_{k \rightarrow +\infty} \int_{\mathbb{R}^n} \varphi \cdot u_k dx = \int_{\mathbb{R}^n} \varphi \cdot u dx = \int_{\mathbb{R}^n} f \operatorname{div}_{\text{NL}}^\alpha(g, \varphi) dx = \int_{\mathbb{R}^n} g \operatorname{div}_{\text{NL}}^\alpha(f, \varphi) dx \quad (4.6)$$

for all $\varphi \in C_c^\infty(\mathbb{R}^n; \mathbb{R}^n)$. In view of (4.3) and (4.6), we must have that

$$u = \nabla_{\text{NL}, w}^\alpha(f, g)$$

as in Definition 4.5, with

$$\|\nabla_{\text{NL}, w}^\alpha(f, g)\|_{L^p(\mathbb{R}^n; \mathbb{R}^n)} \leq C \left(\|\nabla^\alpha f\|_{L^p(\mathbb{R}^n; \mathbb{R}^n)} \|g\|_{L^\infty(\mathbb{R}^n)} + \|f\|_{L^\infty(\mathbb{R}^n)} \|\nabla^\alpha g\|_{L^p(\mathbb{R}^n; \mathbb{R}^n)} \right).$$

Therefore, we can integrate (4.2) against an arbitrary $\varphi \in C_c^\infty(\mathbb{R}^n; \mathbb{R}^n)$ and pass to the limit as $k \rightarrow +\infty$ to get that

$$\int_{\mathbb{R}^n} f g \operatorname{div}^\alpha \varphi dx = \int_{\mathbb{R}^n} \varphi \cdot (g \nabla^\alpha f + f \nabla^\alpha g + \nabla_{\text{NL}, w}^\alpha(f, g)) dx$$

and the conclusion follows. $\square$

Remark 4.6 (The case $p = 1$ in Theorem 1.3). Concerning the validity of Theorem 1.3 in the case $p = 1$, we can make the following observation. If $f, g \in S^{\alpha, 1}(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$, then

$$f g \in S^{\alpha, 1}(\mathbb{R}^n) \iff \exists \nabla_{\text{NL}, w}^\alpha(f, g) \in L^1(\mathbb{R}^n; \mathbb{R}^n), \quad (4.7)$$

in which case

$$\nabla^\alpha(fg) = g \nabla^\alpha f + f \nabla^\alpha g + \nabla_{\text{NL}, w}^\alpha(f, g) \quad \text{in } L^1(\mathbb{R}^n; \mathbb{R}^n). \quad (4.8)$$

Indeed, for a given $h \in C_c^\infty(\mathbb{R}^n)$, by Theorem 3.1 we have $gh \in S^{\alpha, 1}(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$ with

$$\nabla^\alpha(gh) = h \nabla^\alpha g + g \nabla^\alpha h + \nabla_{\text{NL}}^\alpha(g, h) \quad \text{in } L^1(\mathbb{R}^n; \mathbb{R}^n).$$

Therefore, thanks to Proposition 3.10, we can write

$$\begin{aligned} \int_{\mathbb{R}^n} f g \nabla^\alpha h dx &= \int_{\mathbb{R}^n} f \nabla^\alpha(gh) dx - \int_{\mathbb{R}^n} f h \nabla^\alpha g dx - \int_{\mathbb{R}^n} f \nabla_{\text{NL}}^\alpha(g, h) dx \\ &= - \int_{\mathbb{R}^n} g h \nabla^\alpha f dx - \int_{\mathbb{R}^n} f h \nabla^\alpha g dx - \int_{\mathbb{R}^n} f \nabla_{\text{NL}}^\alpha(g, h) dx \end{aligned}$$

for all $h \in C_c^\infty(\mathbb{R}^n)$ and the validity of (4.7) and (4.8) immediately follows. However, in contrast with what happens for the case $p \in (1, +\infty)$ in Theorem 1.3, we do not know if the implication

$$f, g \in S^{\alpha, 1}(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n) \implies \exists \nabla_{\text{NL}, w}^\alpha(f, g) \in L^1(\mathbb{R}^n; \mathbb{R}^n)$$

Figure 1: Remark 4.6 and the open implication in the source paper [1], PDF page 30.